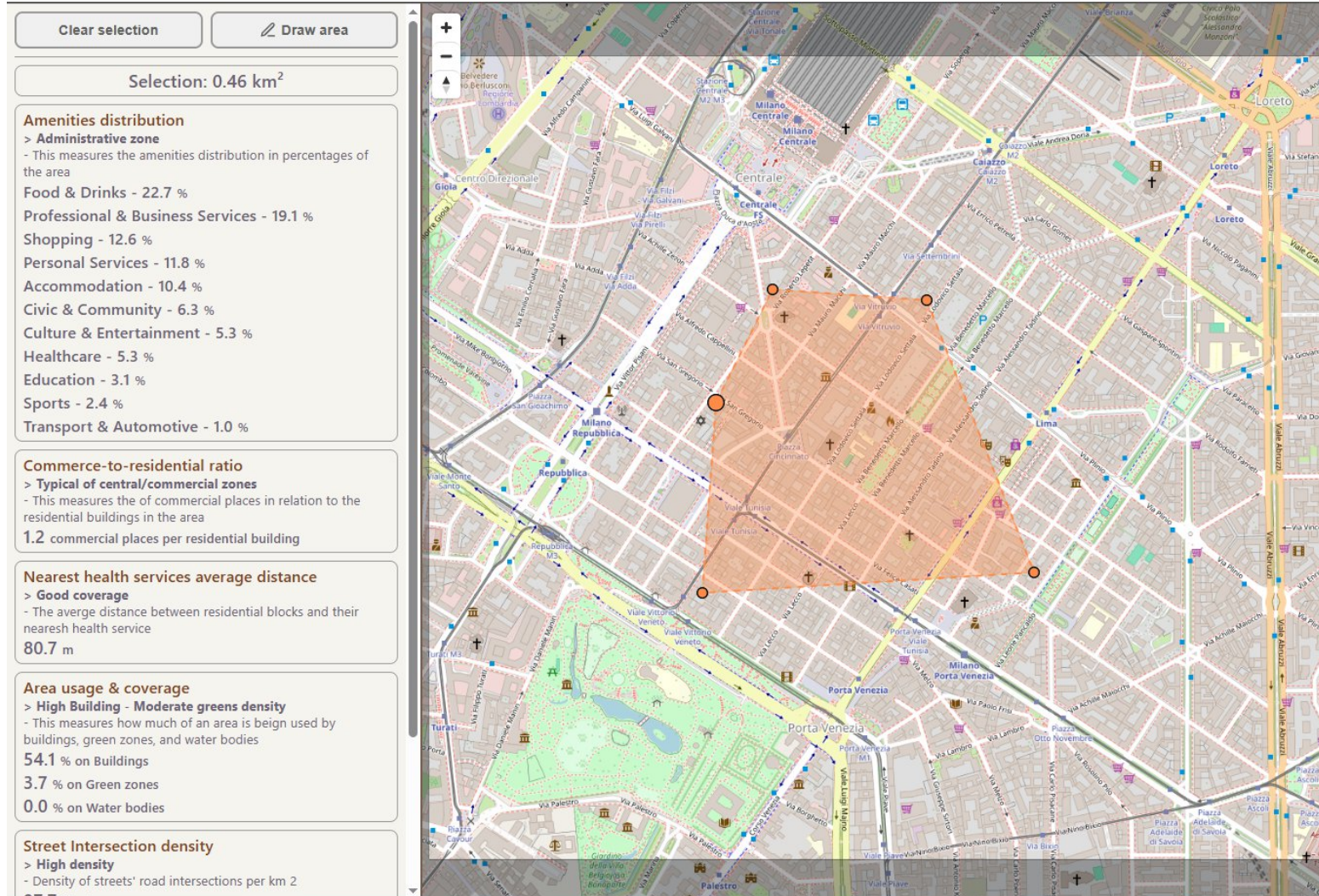


Urban KPI Dashboard — Milan

A data-driven reading of Loreto / Centrale / Porta Venezia



What does this place look like through the KPIs?

- The health coverage of the zone is very high
- Always had the sensation that, moving around milan is slow, now the data back it up
- The area has a lower green zones index than expected and the open space seem to be mostly roads

The place

Why this area?



Milan was selected as the study city for some reasons, the other 2 options were Barcelona or Pisa.

Each city could have a potential set of interesting data, but decided to pick Milan for being more an administrative and industrial than tourism city, for the same reason, picked this part of the district

Selected area: Loreto / Centrale / Porta Venezia.

Interesting mix of landmarks and urban functions.

Large train station

Shopping-focused street

Park with a lake

Dashboard reading

The numbers should tell a story, not just decorate the map

Commerce-to-residential ratio

> Typical of central/commercial zones

- This measures the of commercial places in relation to the residential buildings in the area

1.2 commercial places per residential building

Nearest health services average distance

> Good coverage

- The average distance between residential blocks and their nearest health service

80.7 m

Public transport coverage

> Low coverage

- This measures the coverage % of transport network access stepped by distance

1.0 % at 100m

6.2 % at 250m

24.7 % at 500m

Street Intersection density

> High density

- Density of streets' road intersections per km²

140.8 /km²

The city has data of being a heavy industrialized place, keeping high some amenities, and trading off with others

Life of the data

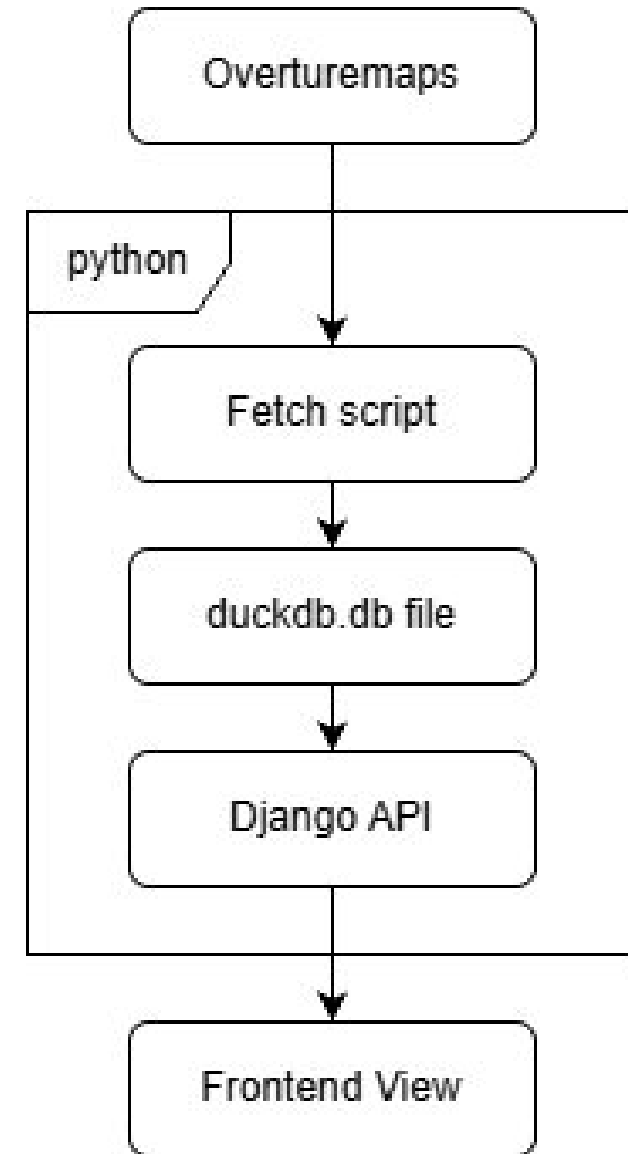
The data is being fetched from a python script directly from overture's map S3 urls, fetching the themes from overturemaps: infrastructure, segment, connector, address, place, building, land, land_use, and water.

When downloaded DuckDB python API stores the data inside tables in-storage outputting a **.db** file,

Then Django server can connect to the generated tables, and create a taxonomy mapper for some categories in one of the KPIs.

Now using DuckDB for making queries, a central module controls which kpis will be launched, each one have its own query that fetches data from the relevant tables.

At last the server receives requests sent when an area is selected, with a set of coords vertexes, using REST requests, the server makes the sequence of queries, and then the frontend displays the information.



Why these KPIs?

- **Area usage and coverage**

Was the first KPI of the pack, the idea behind it is know the amount of free space and green zones that could be used by city inhabitants

- **Amenities distribution**

The idea is to understand what types of amenities are available in the area and how they are distributed. This helps identify the functional character of a neighborhood and whether its amenities are concentrated around particular zones.

- **Commerce-to-residential buildings ratio**

The idea is to measure the relationship and density of commercial activity and residential presence. This helps distinguish areas that are primarily residential from areas with a stronger commercial function.

- **Nearest average distance for residential blocks**

The idea is to measure how geographically accessible healthcare services are to residents. By calculating the average distance from residential buildings to their nearest health service provider, the KPI provides an indication of local healthcare accessibility.

- **Streets intersection density**

Its purpose is to measure the connectivity of the street network. A higher density of intersections generally indicates a more fine-grained network with more route choices and potentially greater pedestrian connectivity.

- **Public transport coverage**

The idea is to measure how much of the area is located within a distance of a public transport access point. Using multiple distances makes it possible to see how coverage changes as the acceptable walking distance increases.

Thresholds & evidence

Street Intersection density

> High density

- Density of streets' road intersections per km²

140.8 /km²

Density	Interpretation
<98/km ²	Below study reference
98–122/km ²	Meets first study threshold
122–334/km ²	Within study-derived range
334–356/km ²	Upper study-derived range
>356/km ²	Above study-derived range

Public transport coverage

> Low coverage

- This measures the coverage % of transport network access stepped by distance

1.0 % at 100m

6.2 % at 250m

24.7 % at 500m

Using as a base, the street road intersections KPI,

There is a IPEN Adult Study that examined the relationship between street-network characteristics and walking, across 14 cities in 10 countries.

with valued threshold varying from 98 intersections/km², that incited walking as a transport method.

This Milan zone has around 150 intersections/km², putting it on the high side of the balance, that may have sense with another KPI, about a low transport coverage in near distance

Why this threshold was retained?

It seemed very interesting, and with data found, may be a valuable indicator

However, the source don't have information about traffic, as it also would be interesting to match

Decisions rejected

Some of the KPIs were not included by 3 main reasons:

- Time constraints
- Not enough relevant data,
 - Beign on of the rejected, A KPI for amount of infrastructure weather-covered walkable-space, or crossroads visibility

Some other rejected KPIs were:

Cycloway distance share in relation to roads

Median building block Size

A KPI remained althought is not that useful as data, it is the building area coverage, the data I can get from it, is not much, and it remained as the control set for the app.

Area usage & coverage
> **High Building - Moderate greens density**
- This measures how much of an area is beign used by buildings, green zones, and water bodies
41.8 % on Buildings
9.6 % on Green zones
0.3 % on Water bodies

Another notorius KPI is the amenities distribution, it is a valuable set of data, but the way it was set, makes it very difficult to put on appropriate thresholds

Amenities distribution
> **Administrative zone**
- This measures the amenities distribution in percentages of the area
Food & Drinks - 23.2 %
Professional & Business Services - 21.1 %
Shopping - 12.6 %
Personal Services - 8.7 %
Civic & Community - 7.5 %
Healthcare - 6.8 %
Accommodation - 6.5 %
Culture & Entertainment - 4.4 %
Transport & Automotive - 3.6 %
Education - 2.9 %
Sports - 2.7 %

What If?

What breaks at scale

10× larger area / 90-second computation

The dataset even beign small and relatively easy to handle, has its problems, with arbitrary definitions or missing categorizations.

With a higher scale data volume, the process would be slower, the best in case of scaling up is:

- Adding event queues
- Splitting the job load in smaller datasets for stepped process
- Using more efficient methods for matching data

One more week

With one more week, i could be able to make some better designed thresholds, more specialized KPIs and a deeper understanding on GeoData in general

Add the posibility to increase or change the district for comparisons between districts, or even cities

From a development point of view, adding features and a better design for both front and back ends would be on the roadmap